



Formula 99 (AFCO 7224)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations
Revision Date: 02/04/2025

Version: 1.1

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture

Product Name: Formula 99

Product Code: AFCO 7224

Intended Use of the Product

Use of the Substance/Mixture: Non-acid bathroom cleaner. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

AFCO
550 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Skin Irrit. 2 H315

Eye Dam. 1 H318

Aquatic Acute 3 H402

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)

:



GHS05

Signal Word (GHS-US)

: Danger

Hazard Statements (GHS-US)

: H315 - Causes skin irritation
H318 - Causes serious eye damage
H402 - Harmful to aquatic life

Precautionary Statements (GHS-US)

: P264 - Wash hands, forearms, and exposed areas thoroughly after handling.
P273 - Avoid release to the environment.
P280 - Wear eye protection, face protection, protective clothing, protective gloves.
P302+P352 - IF ON SKIN: Wash with plenty of soap and water.
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 - Immediately call a POISON CENTER or doctor/physician.
P321 - Specific treatment (see Section 4).
P332+P313 - If skin irritation occurs, get medical advice/attention.
P362 - Take off contaminated clothing and wash before reuse.
P501 - Dispose of contents/container according to local, regional, national, territorial, provincial, and international regulations.

Other Hazards

Other Hazards Not Contributing to the Classification: Exposure may aggravate those with pre-existing eye, skin, or respiratory conditions.

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Unknown Acute Toxicity (GHS-US)

Less than 1 percent of the mixture consists of ingredient(s) of unknown acute toxicity.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Name	Product identifier	% (w/w)	Classification (GHS-US)
Water	(CAS No) 7732-18-5	80-90	Not classified
Tetrasodium EDTA	(CAS No) 64-02-8	5-10	Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Inhalation: dust, mist), H332 Eye Dam. 1, H318 Aquatic Acute 2, H401
2-Butoxyethanol	(CAS No) 111-76-2	1 - 5	Flam. Liq. 4, H227 Acute Tox. 4 (Oral), H302 Acute Tox. 3 (Dermal), H311 Acute Tox. 3 (Inhalation: vapour), H331 Skin Irrit. 2, H315 Eye Irrit. 2A, H319
Sodium metasilicate	(CAS No) 6834-92-0	1 - 5	Met. Corr. 1, H290 Acute Tox. 4 (Oral), H302 Skin Corr. 1B, H314 Eye Dam. 1, H318 STOT SE 3, H335
Alcohols, C12-15, ethoxylated	(CAS No) 68131-39-5	0.1 – 1.0	Eye Dam. 1, H318 Aquatic Acute 1, H400 Aquatic Chronic 3, H412
Poly(oxy-1,2-ethanediyl), .alpha.-hydro-.omega.-hydroxy-, mono-C8-10-alkyl ethers, phosphates	(CAS No) 68130-47-2	0.1 – 1.0	Skin Corr. 1C, H314 Eye Dam. 1, H318
Triethanolamine	(CAS No.) 102-71-6	0.1 – 1.0	Eye Irrit. 2A, H319

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: Remove to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER/doctor/physician if you feel unwell.

Skin Contact: Remove contaminated clothing. Drench affected area with water for several minutes. Immediately call a POISON CENTER or doctor/physician. Wash contaminated clothing before reuse.

Eye Contact: Rinse cautiously with water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/physician.

Ingestion: Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

Most Important Symptoms and Effects Both Acute and Delayed

General: Causes skin irritation and serious eye damage.

Inhalation: None.

Skin Contact: Causes irritation.

Eye Contact: Causes serious eye damage.

Ingestion: Ingestion is likely to be harmful or have adverse effects.

Chronic Symptoms: None known.

Indication of Any Immediate Medical Attention and Special Treatment Needed

If you feel unwell, seek medical advice (show the label where possible).

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SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Use extinguishing media appropriate for surrounding fire.

Unsuitable Extinguishing Media: Do not use a heavy water stream. A heavy water stream may spread burning liquid.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive.

Reactivity: Thermal decomposition generates corrosive vapors.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire. Under fire conditions, hazardous fumes will be present.

Firefighting Instructions: Use water spray or fog for cooling exposed containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: Corrosive vapors. Carbon oxides (CO, CO₂), phosphorus oxides, nitrogen oxides.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid all contact with skin and eyes.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: None.

Environmental Precautions

Prevent entry to sewers and public waters.

Methods and Material for Containment and Cleaning Up

For Containment: Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

Methods for Cleaning Up: Clear up spills immediately and dispose of waste safely. Contact competent authorities after a spill.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash contaminated clothing before reuse.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store in a dry, cool and well-ventilated place. Keep/Store away from extremely low temperatures and incompatible materials. Keep container closed when not in use.

Incompatible Materials: Strong oxidizers, bleach.

Specific End Use(s)

Non-acid bathroom cleaner. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

2-Butoxyethanol (111-76-2)		
Mexico	OEL TWA (mg/m ³)	120 mg/m ³
Mexico	OEL TWA (ppm)	26 ppm
Mexico	OEL STEL (mg/m ³)	360 mg/m ³
Mexico	OEL STEL (ppm)	75 ppm

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USA ACGIH	ACGIH TWA (ppm)	20 ppm
USA OSHA	OSHA PEL (TWA) (mg/m ³)	240 mg/m ³
USA OSHA	OSHA PEL (TWA) (ppm)	50 ppm
USA NIOSH	NIOSH REL (TWA) (mg/m ³)	24 mg/m ³
USA NIOSH	NIOSH REL (TWA) (ppm)	5 ppm
USA IDLH	US IDLH (ppm)	700 ppm
Ontario	OEL TWA (ppm)	20 ppm
Québec	VEMP (mg/m ³)	97 mg/m ³
Québec	VEMP (ppm)	20 ppm

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed.

Personal Protective Equipment: Protective goggles. Gloves.



Materials for Protective Clothing: None usually needed.

Hand Protection: Wear chemically resistant protective gloves.

Eye Protection: Chemical goggles.

Skin and Body Protection: None usually needed.

Respiratory Protection: None usually needed.

Environmental Exposure Controls: None.

Consumer Exposure Controls: Do not eat, drink or smoke during use.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid
Appearance	: Clear, colorless
Odor	: Lemon
Odor Threshold	: Not available
pH (as is)	: 11.4
Relative Evaporation Rate (butylacetate=1)	: Not available
Melting Point	: Not available
Freezing Point	: Not available
Boiling Point	: 103°C (217°F)
Flash Point	: None
Auto-ignition Temperature	: Not available
Decomposition Temperature	: Not available
Flammability (solid, gas)	: Not flammable
Lower Flammable Limit	: Not available
Upper Flammable Limit	: Not available
Vapor Pressure	: Not available
Relative Vapor Density at 20°C	: Not available
Relative Density	: Not available
Specific Gravity	: 1.05
Solubility	: Water: Complete
Partition coefficient: n-octanol/water	: Not available
Viscosity	: Not available
Explosion Data – Sensitivity to Mechanical Impact	: Not expected to present an explosion hazard due to mechanical impact.

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Explosion Data – Sensitivity to Static Discharge : Not expected to present an explosion hazard due to static discharge.

SECTION 10: STABILITY AND REACTIVITY

Reactivity: Thermal decomposition generates corrosive vapors.

Chemical Stability: Stable under recommended handling and storage conditions (see section 7).

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Extremely low temperatures and incompatible materials.

Incompatible Materials: Strong oxidizers, bleach.

Hazardous Decomposition Products: Carbon oxides (CO, CO₂), phosphorus oxides, nitrogen oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Acute Toxicity: Not classified.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: Causes skin irritation. (pH: 11.4)

Serious Eye Damage/Irritation: Causes serious eye damage. (pH: 11.4)

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not classified.

Carcinogenicity: Not classified.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): Not classified.

Aspiration Hazard: Not classified.

Symptoms/Injuries After Inhalation: May cause respiratory irritation.

Symptoms/Injuries After Skin Contact: Causes irritation.

Symptoms/Injuries After Eye Contact: Causes serious eye damage.

Symptoms/Injuries After Ingestion: Ingestion is likely to be harmful or have adverse effects.

Chronic Symptoms: None known.

Information on Toxicological Effects - Ingredient(s)

LD50 and LC50 Data:

Sodium metasilicate (6834-92-0)	
LD50 Oral Rat	600 mg/kg
Alcohols, C12-15, ethoxylated (68131-39-5)	
LD50 Oral Rat	1600 mg/kg
LD50 Dermal Rabbit	2500 mg/kg
Tetrasodium EDTA (64-02-8)	
LD50 Oral Rat	1780 mg/kg
ATE US (dust, mist)	1.50 mg/l/4h
Triethanolamine (102-71-6)	
LD50 Oral Rat	6400 mg/kg
2-Butoxyethanol (111-76-2)	
LD50 Oral Rat	470 mg/kg
LD50 Dermal Rat	220 mg/kg
LC50 Inhalation Rat	450 ppm/4h
ATE US (vapors)	3.84 mg/l/4h
IARC Group	3
National Toxicity Program (NTP) Status	Evidence of Carcinogenicity.

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SECTION 12: ECOLOGICAL INFORMATION

Toxicity

Ecology - General: This material is hazardous to the aquatic environment. Keep out of sewers and waterways.

Ecology - Water: Harmful to aquatic life.

Tetrasodium EDTA (64-02-8)	
LC50 Fish 1	41 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
LC50 Fish 2	59.8 mg/l (Exposure time: 96 h - Species: Pimephales promelas [static])
ErC50 (algae)	2.77 mg/l (72 h - Species: Desmodesmus subspicatus)

2-Butoxyethanol (111-76-2)	
LC50 Fish 1	1490 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
EC50 Daphnia 1	1000 mg/l (Exposure time: 48 h - Species: Daphnia magna)
LC50 Fish 2	2950 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus)

Sodium metasilicate (6834-92-0)	
LC50 Fish 1	210 mg/l (Exposure time: 96 h - Species: Brachydanio rerio [semi-static])
LC50 Fish 2	210 mg/l (Exposure time: 96 h - Species: Brachydanio rerio)

Alcohols, C12-15, ethoxylated (68131-39-5)	
LC50 Fish 1	0.59 mg/l

Triethanolamine (102-71-6)	
LC50 Fish 1	5600 mg/l (Exposure time: 96 h - Species: Pimephales promelas)
EC50 Daphnia 1	947 mg/l (Exposure time: 48 h - Species: Daphnia magna)

Persistence and Degradability: Not available.

Bioaccumulative Potential

Tetrasodium EDTA (64-02-8)	
Log Pow	5.01 (calculated)

2-Butoxyethanol (111-76-2)	
Log Pow	0.81 (at 25°C)

Mobility in Soil Not available.

Other Adverse Effects Not available.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

Ecology – Waste Materials: Avoid release to the environment.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT

Proper Shipping Name : CORROSIVE LIQUID, BASIC , ORGANIC , N.O.S. (Tetrasodium EDTA)
Hazard Class : 8
Identification Number : UN3267
Label Codes : 8
Packing Group : III
ERG Number : 153



14.2 In Accordance with IMDG

Proper Shipping Name : CORROSIVE LIQUID, BASIC , ORGANIC , N.O.S. (Tetrasodium EDTA)
Hazard Class : 8
Identification Number : UN3267
Packing Group : III
Label Codes : 8
EmS-No. (Fire) : F-A
EmS-No. (Spillage) : S-B



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14.3 In Accordance with IATA

Proper Shipping Name : CORROSIVE LIQUID, BASIC , ORGANIC , N.O.S. (Tetrasodium EDTA)
Packing Group : III
Identification Number : UN3267
Hazard Class : 8
Label Codes : 8
ERG Code (IATA) : 8L



14.4 In Accordance with TDG

Proper Shipping Name : CORROSIVE LIQUID, BASIC , ORGANIC , N.O.S. (Tetrasodium EDTA)
Packing Group : III
Hazard Class : 8
Identification Number : UN3267
Label Codes : 8



***Note:** AF7224X (12 x 1 QT) is allowed to ship as Limited Quantity.

SECTION 15: REGULATORY INFORMATION

US Federal Regulations

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SARA Section 311/312 Hazard Classes	Immediate (acute) health hazard. Delayed (chronic) health hazard.
Water (7732-18-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Tetrasodium EDTA (64-02-8)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
2-Butoxyethanol (111-76-2)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Sodium metasilicate (6834-92-0)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Alcohols, C12-15, ethoxylated (68131-39-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Poly(oxy-1,2-ethanediyl), .alpha.-hydro.-omega.-hydroxy-, mono-C8-10-alkyl ethers, phosphates (68130-47-2)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Triethanolamine (102-71-6)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	

US State Regulations

2-Butoxyethanol (111-76-2)	
U.S. - California - SCAQMD - Toxic Air Contaminants - Non-Cancer Acute U.S. - California - Toxic Air Contaminant List (AB 1807, AB 2728) RTK - U.S. - New Jersey - Right to Know Hazardous Substance List U.S. - New Jersey - Special Health Hazards Substances List U.S. - New York - Occupational Exposure Limits - Skin Designations U.S. - New York - Occupational Exposure Limits - TWAs RTK - U.S. - Pennsylvania - RTK (Right to Know) List U.S. - Texas - Effects Screening Levels - Long Term U.S. - Texas - Effects Screening Levels - Short Term	
Sodium metasilicate (6834-92-0)	
U.S. - Texas - Effects Screening Levels - Long Term U.S. - Texas - Effects Screening Levels - Short Term	

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Tetrasodium EDTA (64-02-8)
U.S. - Texas - Effects Screening Levels - Long Term
U.S. - Texas - Effects Screening Levels - Short Term
Alcohols, C12-15, ethoxylated (68131-39-5)
U.S. - Texas - Effects Screening Levels - Long Term
U.S. - Texas - Effects Screening Levels - Short Term

Canadian Regulations

Water (7732-18-5)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Tetrasodium EDTA (64-02-8)
Listed on the Canadian DSL (Domestic Substances List) inventory.
2-Butoxyethanol (111-76-2)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Sodium metasilicate (6834-92-0)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Alcohols, C12-15, ethoxylated (68131-39-5)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Poly(oxy-1,2-ethanediyl), .alpha.-hydro-.omega.-hydroxy-, mono-C8-10-alkyl ethers, phosphates (68130-47-2)
Listed on the Canadian DSL (Domestic Substances List) inventory.
Triethanolamine (102-71-6)
Listed on the Canadian DSL (Domestic Substances List) inventory.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date	:	02/04/2025
Other Information	:	This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

GHS Full Text Phrases:

Acute Tox. 3 (Dermal)	Acute toxicity (Dermal) Category 3
Acute Tox. 3 (Inhalation: dust, mist)	Acute toxicity (inhalation: dust, mist) Category 3
Acute Tox. 4 (Inhalation: dust, mist)	Acute toxicity (inhalation: dust, mist) Category 4
Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4
Aquatic Acute 1	Hazardous to the aquatic environment - Acute Hazard Category 1
Aquatic Acute 2	Hazardous to the aquatic environment - Acute Hazard Category 2
Aquatic Chronic 3	Hazardous to the aquatic environment - Chronic Hazard Category 3
Skin Corr. 1B	Skin corrosion/irritation Category 1B
Skin Corr. 1C	Skin corrosion/irritation Category 1C
Skin Irrit. 2	Causes skin corrosion
Eye Dam. 1	Serious eye damage/eye irritation Category 1
Eye Irrit. 2A	Serious eye damage/eye irritation Category 2A
Flam. Liq. 4	Flammable liquids Category 4
Met. Corr. 1	Corrosive to metals Category 1
STOT SE 3	Specific target organ toxicity (single exposure) Category 3
H227	Combustible liquid
H290	May be corrosive to metals
H302	Harmful if swallowed
H311	Toxic in contact with skin
H314	Causes severe skin burns and eye damage

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H315	Causes skin irritation
H318	Causes serious eye damage
H319	Causes serious eye irritation
H331	Toxic if inhaled
H332	Harmful if inhaled
H335	May cause respiratory irritation
H400	Very toxic to aquatic life
H401	Toxic to aquatic life
H412	Harmful to aquatic life with long lasting effects

NFPA Health Hazard

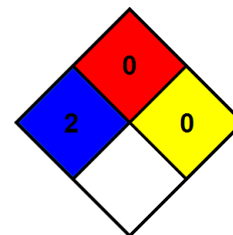
: 2 - Intense or continued exposure could cause temporary incapacitation or possible residual injury unless prompt medical attention is given.

NFPA Fire Hazard

: 0 - Materials that will not burn.

NFPA Reactivity

: 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.



HMIS III Rating

Health

: 2 Moderate Hazard - Temporary or minor injury may occur.

Flammability

: 0 - Minimal Hazard.

Physical

: 0 - Minimal Hazard.

Party Responsible for the Preparation of This Document

AFCO

550 Development Avenue

Chambersburg, PA 17201

T: 800-345-1329

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)